학술 검색 세션의 탐색 맥락 구조화를 위한 대규모 로그 기반 탐색 행동 분석 및 지식그래프 구축

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Motivation

이 논문은 정부(교육부)의 재원으로 한국산업기술진흥원의 지원(P0030395, 첨단산업인재양성부트캠프)과 2026 정부(과학기술정보통신부)의 지원을 받아 수행된 연구임 (RS-2026-25490215)

Background & Challenges

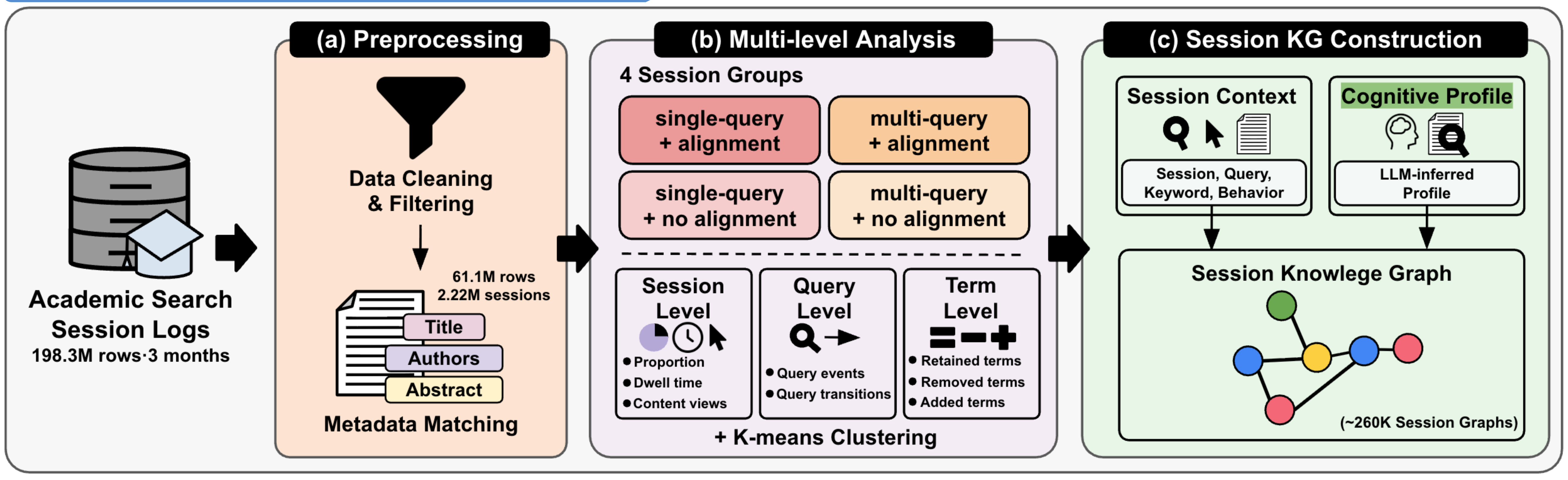
- Understanding **search context** is important for supporting users in academic search
- Search settings **without user authentication** make it difficult to link **search histories across sessions**
- **Limited access to personal histories or user profiles**

The Need for Session-level Context Modeling

- An approach that captures **search context from information observable within individual sessions**, without relying on **cross-session user histories or profiles**
- A structured representation of **session-level search context** for supporting academic search **without user authentication**

We propose a session-level context modeling approach that builds on **large-scale academic search log analysis** and structures **search context** and **inferred cognitive states** as a **knowledge graph**

Methodology



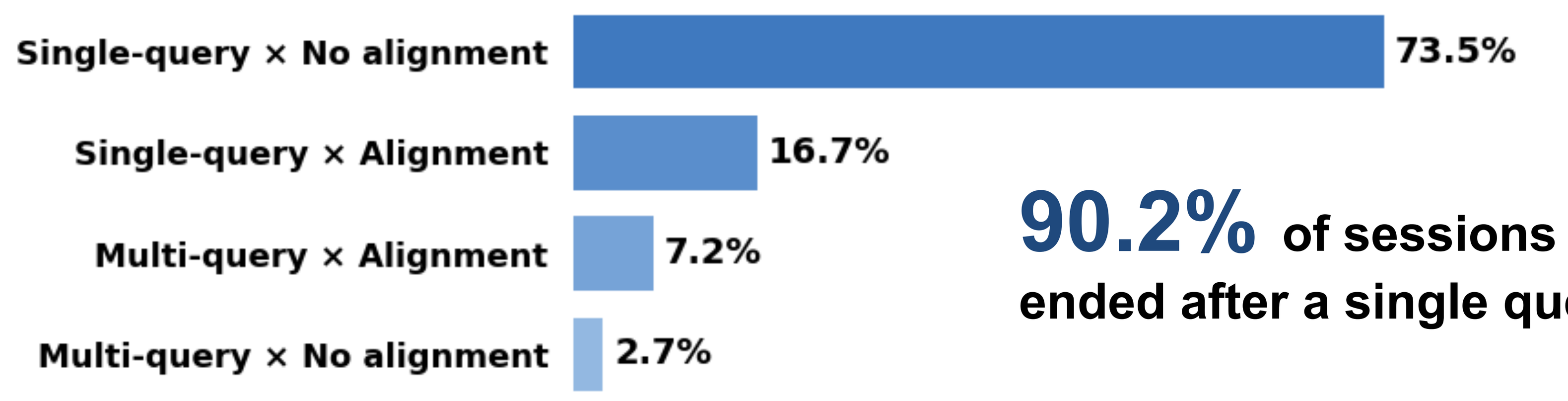
* **Alignment behaviors:** full-text access, citation, export, save, and share

- **(a) Preprocessing:** Cleaning and filtering large-scale academic search logs and matching them with content metadata.
- **(b) Multi-level Analysis:** Grouping sessions by query count × alignment* and analyzing search patterns with K-means clustering.
- **(c) Session KG Construction:** Structuring session context and LLM-inferred cognitive profiles as session knowledge graphs.

Results

Large-scale Search Behavior Analysis

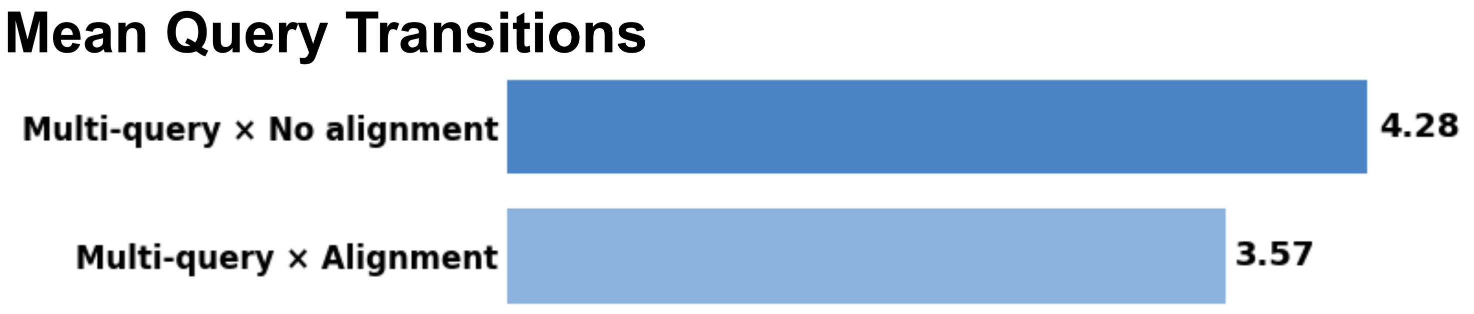
Session-level Search Patterns



90.2% of sessions ended after a single query

Session Group	Dwell Time	Content Use
Single-query × No alignment	249.6s	2.01
Single-query × Alignment	983.0s	2.97
Multi-query × No alignment	1,282.8s	11.30
Multi-query × Alignment	1,923.7s	6.49

Mean Query Transitions



Search context should be inferred **beyond queries**, incorporating **interaction** and **content-use signals**

K-means Clustering

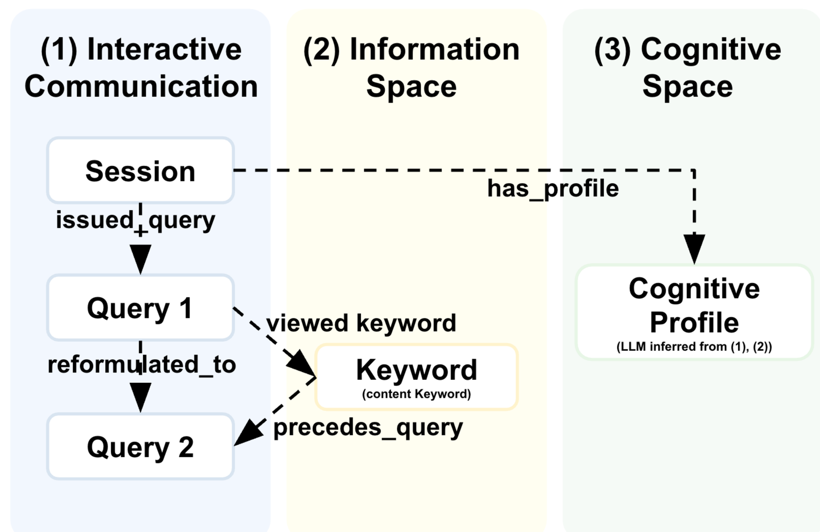
Example Clustering Result: Single-query × Alignment

Brief Exploration (43.41%)	Active Exploration (40.10%)	Intensive Inspection (8.86%)	Long-stay Exploration (7.62%)
1.5 contents viewed · 14.5 interactions · alignment within 70s	3.1 contents viewed · 26.8 interactions · alignment within 133s	5.2 contents viewed · 65.9 interactions · alignment within 573s	3.6 contents viewed · 36.4 interactions · alignment within 1,644s

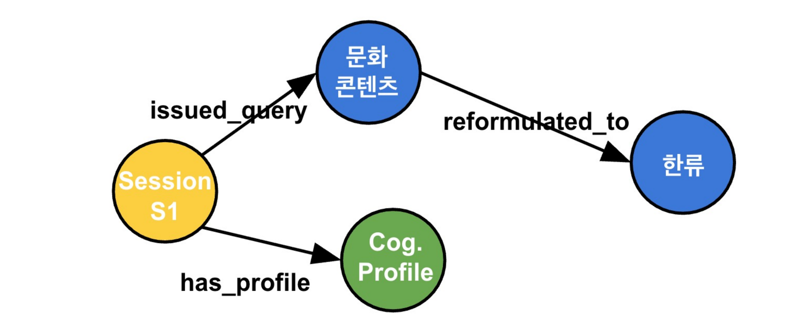
Heterogeneous search patterns indicate the need to represent **session-specific search context** and **cognitive state**

Session Knowledge Graph Construction

Three-space Ontology



Example Session Graph



Cognitive Profile

The user is exploring the concept of '한류' (Hallyu) within cultural content, indicating a focused interest. They exhibit a lateral shift in queries, suggesting a desire for deeper understanding, ... indicating possible uncertainty or a need for confirmation.

Constructed session knowledge graphs for **~263K sessions** using **query, behavior, content, and cognitive profiles**

Preliminary Evaluation

Task: *Context-aware next-query prediction* · ***Data:*** *12K-session pool, 600 test instances*

Metric	Popularity	Query Repeat	Query−only (q_t)	Session KG
Hit@5	0.113	0.158	0.182	**0.205**
Recall@5	0.036	0.083	0.079	**0.104**

Conclusion & Future Work

- Representing search context **w/o user history** through a session KG integrating query, behavior, content, and cognitive signals
- Further evaluate graph-based context-aware recommendation